Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I will use functions, if statements, lists, loops, and dataclasses. My project needs these to store movies, calculate ratings, and show results. I want to improve my skills with lists, dataclasses, and building drafter websites with multiple pages.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

The easiest part will be collecting movie scores and calculating the average rating because I have done similar work before. The hardest part will be keeping track of movies across pages and filtering movies by genre because it uses more complex state management.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

The parts that will take the most time are creating multiple pages, storing movie data, and adding filters and statistics. I'll first build the movie rating form and average rating calculator, then add the movie list and extra features afterward.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Distractions like my phone and video games could slow me down, so I will work in a quiet place and focus on one task at a time. I will use the course notes, Drafter documentation, class examples, and office hours if I need help. Right now, I do not have any questions about the project requirements.

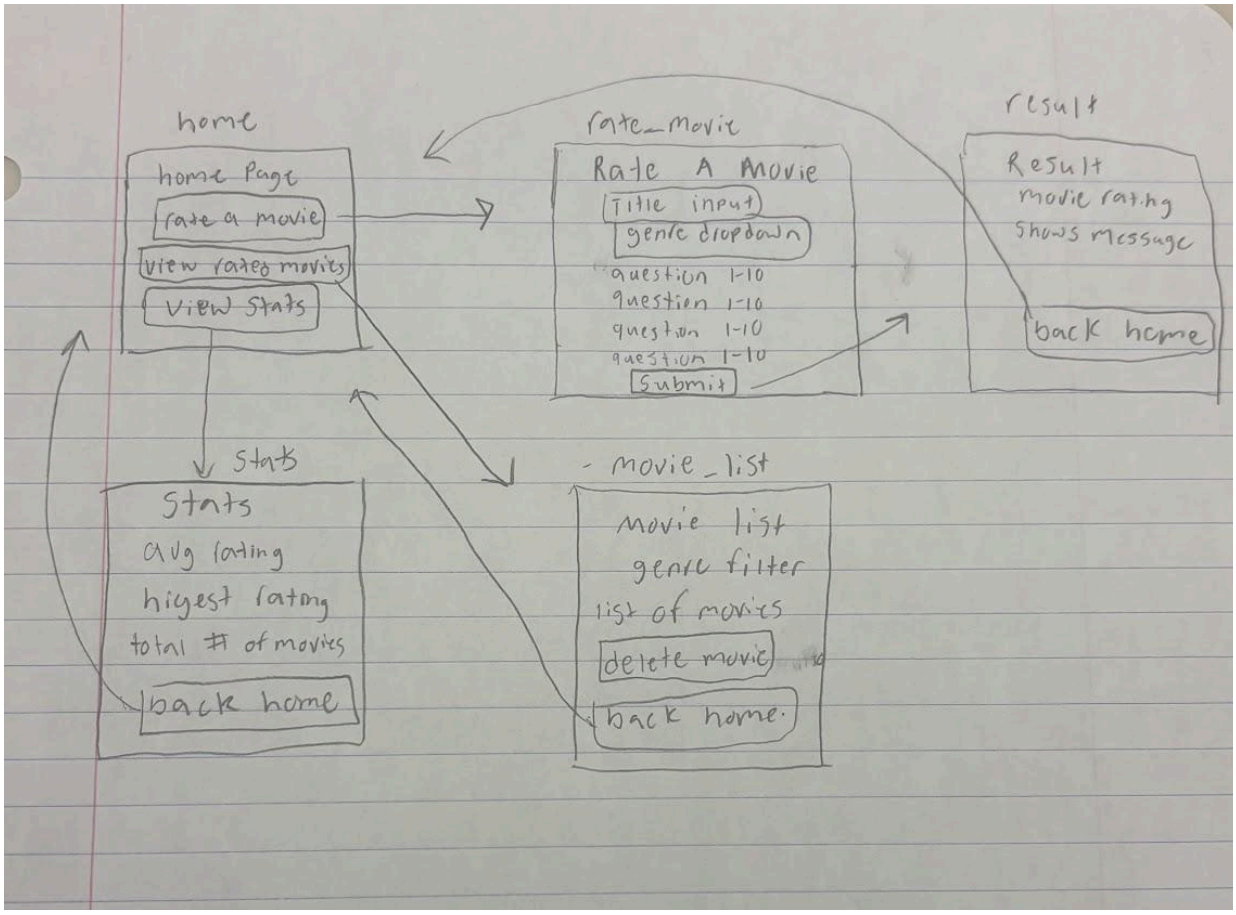
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


home
home Page
rate a movie
view rated movies
View Stats
rate_movie
Rate A Movie
Title input
genre dropdown
question 1-10
question 1-10
question 1-10
question 1-10
Submit
result
Result
movie rating
shows message
back home
stats
Stats
avg rating
higest rating
total # of movies
back home
movie_list
movie list
genre filter
list of movies
delete movie
back home

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	The purpose of my site is to have a list of the movies you've watched and a permanent rating system.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	1. movies: list[Movie] The movies attribute stores all the movies the user has rated. It changes when a movie is added or deleted. 2. filter_genre: str The filter_genre attribute stores the genre the user wants to view. It changes when the user picks a different genre. 3. average_rating: float The average_rating attribute stores the average rating of all saved movies. It updates whenever a movie is added or removed. 4. top_movie: str The top_movie attribute stores the highest-rated movie. It changes when a movie gets a higher rating than the current top movie. Movie Dataclass The Movie dataclass stores information about a single movie, including its title, genre, overall rating, and verdict. A new Movie object is created each time a user rates a movie.
Second Dataclass	Movie dataclass ● Title: str

Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	• Genre: str • Overall_rating: float • Verdict: str A new Movie object is created each time a user rates a movie.
Routes (minimum 7) The names of your route functions	1. home 2. rate_movie 3. submit_rating 4. result 5. movie_list 6. stats 7. delete_movie
Pages (minimum 5) A brief description of the pages in your site	1. Home Page The home page welcomes the user and has buttons to rate a movie, view movies, and view stats. 2. Rate Movie Page The user enters a movie title, genre, and ratings for each category. 3. Results Page This page shows the movie's overall rating and a message like "Must Watch" or "Skip It." 4. Movie List Page This page shows all saved movies and lets the user filter them by genre. 5. Stats Page This page shows the average rating, top-rated movie, and total number of movies rated.
User Input (minimum 3)	1. TextBox (Movie Title)

Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	The user enters the name of the movie they want to rate. 2. SelectBox (Genre) The user chooses the movie genre, such as Action, Comedy, or Horror. 3. SelectBox (Scores 1-10) The user selects scores from 1 to 10 for story, acting, visuals, sound, pacing, and enjoyment. 4. SelectBox (Filter Genre) The user chooses a genre to filter the movie list and only see movies from that genre.
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	1. Rating Verdict An if statement will check the overall rating and show a message like "Must Watch," "Decent," or "Skip It." 2. Top Movie An if statement will check if a movie has a higher rating than the current top movie and update it if needed. 3. Genre Filter An if statement will check if a movie matches the selected genre before showing it in the movie list.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in	A loop will go through the movies list and display every movie the user has rated. This is important because it lets the user see all of their saved movies

your State attribute. Description of this loop and why it is meaningful to the site	and filter them by genre.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Replace this text with your answer. I did not use an LLM

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.